

Signature of sex chromosomes (further analysis)

```
library(tidyverse)
library(tidymodels)
library(readxl)
```

Read and make tx2gene file to replace gene IDs for some species

```
Und_tx2gene <- readr::read_tsv("data/TPM/Und_tx2gene.tsv", col_select = c(1,2), col_names = c("tx", "gene"))
Si_tx2gene <- readr::read_tsv("data/TPM/Schizocladia-ischiensis.tsv", col_select = c(1,2))
```

Read signature master file

```
readxl::excel_sheets("data/dataset_S3_master_tables.xlsx")
```

```
## [1] "Ectocarpus"          "Desmarestia-dudresnayi" "Desmarestia-herbacea"
## [4] "Undaria"            "Scitosyphon"          "Schizocladia"

Ec_sig <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Ectocarpus") |>
  dplyr::select(gene_id, dplyr::contains("Signature"))
Ddud_sig <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Desmarestia-dudresnayi") |>
  dplyr::select(gene_id, dplyr::contains("Signature"))
Dher_sig <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Desmarestia-herbacea") |>
  dplyr::select(gene_id, dplyr::contains("Signature"))
Spro_sig <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Scitosyphon") |>
  dplyr::select(gene_id, dplyr::contains("Signature"))

# species where renaming is needed

Und_sig <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Undaria") |>
  # rename gene_id
  dplyr::left_join(Und_tx2gene) |> dplyr::select(-gene_id) |>
  dplyr::select(gene_id = tx, dplyr::contains("Signature"))
Si_sig <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Schizocladia") |>
  # rename gene_id
  dplyr::left_join(Si_tx2gene) |> dplyr::select(-gene_id) |>
  dplyr::select(gene_id = tx, dplyr::contains("Signature"))
```

Read and process orthogroup data to obtain one-to-one orthologues

```
# filter longest isoform in Ectocarpus
```

```
library(rtracklayer)
```

```
Ec_data.gtf <- rtracklayer::import("data/annotation/Ec32_V5.longest_mRNA.SDR_F.AGAT.mod.gtf")
Ec_data.gtf.df <- Ec_data.gtf |>
  rtracklayer::as.data.frame() |>
  dplyr::select(seqnames, gene_id) |>
  dplyr::distinct()
```

```
Ec_sig.chr <-
  Ec_sig |>
  dplyr::left_join(Ec_data.gtf.df) |>
```

```

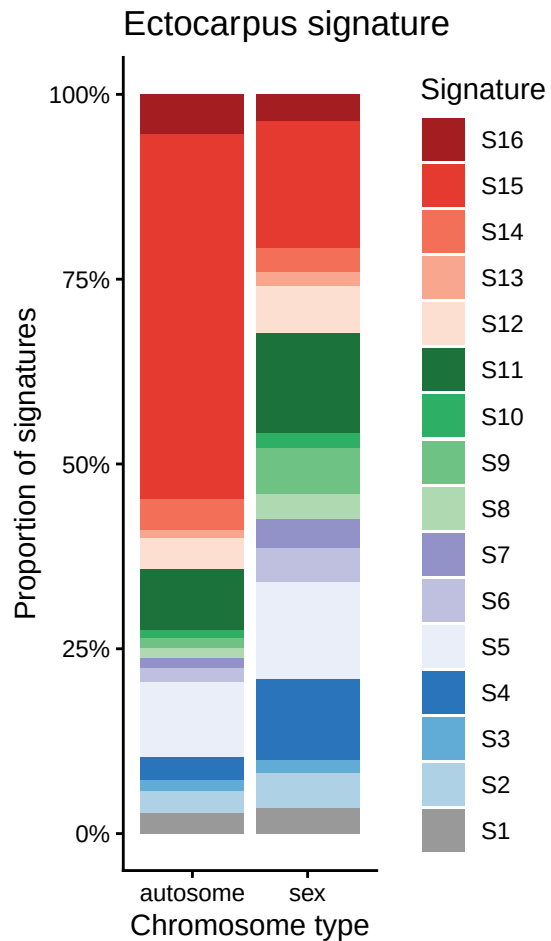
dplyr::mutate(chr_type = dplyr::case_when(
  seqnames == "chr_13" ~ "sex",
  .default = "autosome"
)) |>
dplyr::mutate(Signature.Female)

Ec_sig.chr[Ec_sig.chr == "NA"] <- NA

color_table <-
  tibble::tibble(
    Signature = paste0('S',1:16),
    Colour = c("#a31d21", "#e43a30", "#f36f57", "#f8a68d", "#fddfd1", "#1b733b",
      "#2daf66", "#6ec283", "#aed9b1", "#9392c8", "#bfbfd5", "#e9eef8",
      "#2a74bb", "#61abd7", "#afd1e5", "#999999")
  )

Ec_sig.chr |>
  dplyr::mutate(Signature.Female = factor(Signature.Female, levels = paste0('S',16:1))) |>
  tidyr::drop_na() |>
  ggplot2::ggplot(aes(x = chr_type, fill = Signature.Female)) +
  ggplot2::geom_bar(position="fill") +
  ggplot2::scale_y_continuous(labels = scales::percent) +
  ggplot2::scale_fill_manual(values = setNames(color_table$Colour, color_table$sp)) +
  ggplot2::labs(title = "Ectocarpus signature",
    x = "Chromosome type",
    y = "Proportion of signatures",
    fill = "Signature") +
  ggplot2::theme_classic()

```



```
Ec_sig.chr |>
  dplyr::count(.by = chr_type)

## # A tibble: 2 x 2
##   .by      n
##   <chr>  <int>
## 1 autosome 16179
## 2 sex      434

# we have Ec data already

# Spro chr_13
Spro_data.gtf <- rtracklayer::import("data/annotation/convert_gffread/Scytosiphon-promiscuus_MALE.chr.gtf")
Spro_data.gtf.df <- Spro_data.gtf |>
  rtracklayer::as.data.frame() |>
  dplyr::select(seqnames, gene_id = ID) |>
  dplyr::distinct() |>
  dplyr::mutate(sp = "Spro") |>
  dplyr::mutate(chr_type = dplyr::case_when(
    seqnames == "chr_13" ~ "sex",
    .default = "autosome"
  ))

# Und HiC_scaffold_23
# there are apparently 30 'chromosome-level' scaffolds.
# I confirmed this with seqkit.
```

```

Und_data.gtf <- rtracklayer::import("data/annotation/PUBLIC_Undaria-pinnatifida_MALE.jv3.gtf")
Und_data.gtf.df <- Und_data.gtf |>
  rtracklayer::as.data.frame() |>
  dplyr::select(seqnames, gene_id = transcript_id) |>
  dplyr::distinct() |>
  dplyr::mutate(sp = "Und") |>
  dplyr::mutate(chr_type = dplyr::case_when(
    seqnames == "HiC_scaffold_23" ~ "sex",
    .default = "autosome"
  ))
# Dher chr_03
Dher_data.gtf <- rtracklayer::import("data/annotation/convert_gffread/Desmarestia-herbacea.reference.gtf")
Dher_data.gtf.df <- Dher_data.gtf |>
  rtracklayer::as.data.frame() |>
  dplyr::select(seqnames, gene_id = ID) |>
  dplyr::distinct() |>
  dplyr::mutate(sp = "Dher") |>
  dplyr::mutate(chr_type = dplyr::case_when(
    seqnames == "chr_03" ~ "sex",
    .default = "autosome"
  ))
# Ddud pseudo-chr_03
Ddud_data.gtf <- rtracklayer::import("data/annotation/Desmarestia-dudresnayi.cleaned2025.jv.gffread.gtf")
Ddud_data.gtf.df <- Ddud_data.gtf |>
  rtracklayer::as.data.frame() |>
  dplyr::select(seqnames, gene_id = transcript_id) |>
  dplyr::distinct() |>
  dplyr::mutate(sp = "Ddud") |>
  dplyr::mutate(chr_type = dplyr::case_when(
    seqnames == "chr_03" ~ "pseudo-sex",
    .default = "autosome"
  )) |>
  dplyr::mutate(gene_id = stringr::str_remove(gene_id, "\\..t1$")) # remove isoform suffix

gtf.df.all <-
  Ec_data.gtf.df |>
  dplyr::mutate(sp = "Ec") |>
  dplyr::mutate(chr_type = dplyr::case_when(
    seqnames == "chr_13" ~ "sex",
    .default = "autosome"
  )) |>
  rbind(Spro_data.gtf.df) |>
  rbind(Und_data.gtf.df) |>
  rbind(Dher_data.gtf.df) |>
  rbind(Ddud_data.gtf.df)

sig.all <-
  Ec_sig |> dplyr::select(1, Signature = Signature.Female) |>
  rbind(Ddud_sig) |>
  rbind(Dher_sig |> dplyr::select(1, Signature = Signature.Female)) |>
  rbind(Spro_sig |> dplyr::select(1, Signature = Signature.Female)) |>
  rbind(Und_sig |> dplyr::select(1, Signature = Signature.Female)) |>
  rbind(Si_sig)

```

```

sig.all[sig.all == "NA"] <- NA # replace "NA" with NA

sig.all.gtf.df.all <-
  sig.all |>
  dplyr::left_join(gtf.df.all) |>
  dplyr::mutate(Signature = factor(Signature, levels = paste0('S',16:1)))

rect_df <- sig.all.gtf.df.all |>
  dplyr::distinct(sp) |>
  dplyr::filter(sp == "Ddud") |>
  dplyr::mutate(xmin = -Inf, xmax = Inf, ymin = -Inf, ymax = Inf) |>
  dplyr::mutate(sp = factor(sp, levels = c(
    "Ec", "Spro", "Und", "Dher", "Ddud"
  )))

p_sig.all.gtf.df.all <- sig.all.gtf.df.all |>
  dplyr::mutate(sp = factor(sp, levels = c(
    "Ec", "Spro", "Und", "Dher", "Ddud"
  ))) |>
  tidyr::drop_na()

sig.all.gtf.df.all |>
  tidyr::drop_na() |>
  dplyr::summarise(n_genes = n(), .by = c(sp, chr_type))

## # A tibble: 10 x 3
##   sp    chr_type  n_genes
##   <chr> <chr>      <int>
## 1 Ec    autosome    16179
## 2 Ec    sex          412
## 3 Ddud  autosome    17865
## 4 Ddud  pseudo-sex   368
## 5 Dher  autosome    16024
## 6 Dher  sex          230
## 7 Spro  autosome    18271
## 8 Spro  sex          904
## 9 Und   autosome    22505
## 10 Und  sex          308

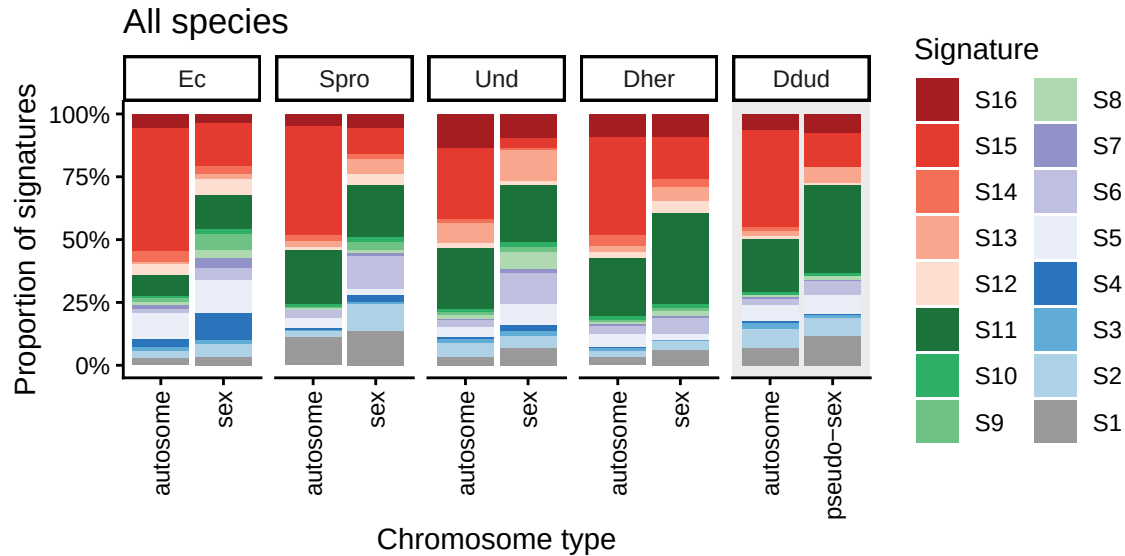
p_sig.all.gtf.df.all |>
  ggplot2::ggplot(aes(x = chr_type, fill = Signature)) +
  # shading
  ggplot2::geom_rect(
    data = rect_df,
    aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
    inherit.aes = FALSE,
    fill = "grey", alpha = 0.3
  ) +
  ggplot2::geom_bar(position="fill") +
  ggplot2::scale_y_continuous(labels = scales::percent) +
  ggplot2::scale_fill_manual(values = setNames(color_table$Colour, color_table$sp)) +
  ggplot2::labs(
    title = "All species",
    x = "Chromosome type",

```

```

y = "Proportion of signatures",
fill = "Signature"
) +
ggplot2::theme_classic() +
ggplot2::facet_grid(. ~ sp, scales = "free_x") +
ggplot2::guides(x = guide_axis(angle = 90),
                fill=guide_legend(ncol=2))

```



```

# add Chi-squared test for independence.
sig.all.gtf.df.all |>
  tidyr::drop_na() |>
  dplyr::group_by(sp) |>
  dplyr::summarise(
    chisq_test = list(
      chisq.test(
        table(chr_type, Signature)
      )
    ),
    .groups = "drop"
  ) |>
  dplyr::mutate(
    p_value = purrr::map_dbl(chisq_test, ~ .$p.value)
  ) |>
  dplyr::select(sp, p_value) |>
  dplyr::mutate(
    p_adj = p.adjust(p_value, method = "BH")
  )

```

```

## # A tibble: 5 x 3
##   sp      p_value      p_adj
##   <chr>    <dbl>    <dbl>
## 1 Ddud  2.33e- 26  2.92e- 26
## 2 Dher  5.70e- 12  5.70e- 12
## 3 Ec    5.28e- 57  1.32e- 56
## 4 Spro  6.00e-204  3.00e-203
## 5 Und   8.45e- 56  1.41e- 55

```